Immune Surveillance and Skin Cancer Risk

Several lines of evidence challenge the interpretation of immune surveillance as the primary mechanism protecting against skin cancer. Cyclophosphamide is a well-known mutagen, and cyclosporine promotes tumor growth in vitro and in immune-deficient mice—organisms that lack an immune system to suppress—suggesting direct carcinogenic effects independent of immunosuppression. Azathioprine acts as a mutagen when combined with UVA irradiation. While HIV patients have a modestly increased incidence of squamous cell carcinoma (SCC), these tumors often occur at sun-shielded sites and are associated with human papillomavirus (HPV) infection.

Skin cancers are frequently reported in patients with leukemia or lymphoma, conditions that involve immunodeficiency. Although many published studies lack controls or detailed patient data, an approximately eightfold increased risk of skin cancer in chronic lymphocytic leukemia appears valid. However, most of these patients have also been exposed to immunosuppressive drugs or radiation, making it difficult to disentangle the effects of immune suppression from the mutagenic effects of prior treatments.

In contrast, Merkel cell carcinoma appears to be genuinely sensitive to immune function. Its incidence increases 10- to 20-fold in solid-organ transplant recipients and approximately 13-fold in HIV patients and those with chronic lymphocytic leukemia (see Chap. 120). The risk of metastasis and mortality is also elevated across all types of immunosuppressed patients. Regardless of the underlying mechanism, close monitoring of immunosuppressed individuals is essential, as early detection allows for curative treatment even in this high-risk population.

Animal Models of Skin Cancer: Clinical Implications

Genetic Risk Factors for Ultraviolet Radiation Carcinogenesis

Pigmentation and Initial Damage

Melanin plays a critical role in protecting the skin against cancer development. The Fitzpatrick skin phototype classification (see Chap. 89), which considers both baseline pigmentation and response to ultraviolet radiation (UVR)—such as tendency to burn or tan—can account for up to a 100-fold difference in skin cancer susceptibility.

Individuals with oculocutaneous albinism, characterized by deficient melanin production due to various genetic mutations, face a markedly increased risk of skin cancer (see Chap. 71). This risk primarily involves non-melanoma skin cancers, though melanoma incidence is also elevated.

Another hereditary risk factor—red or blonde hair—is linked to polymorphisms in the **melanocortin 1 receptor** (MC1R) (see Chap. 70). This G-protein coupled receptor binds melanocyte-stimulating hormone and is central to melanogenesis. Certain loss-of-function mutations result in the production of **pheomelanin** instead of the more photoprotective **eumelanin**.

This distinction is crucial for skin cancer prevention. Pheomelanin is a poor free radical scavenger; upon UVR exposure, it degrades and generates superoxide radicals. Thus, pheomelanin can act as a photosensitizer, contributing to oxidative stress and even inducing apoptosis in neighboring cells. The combination of reduced protection and increased oxidative damage in the epidermis likely contributes to the heightened skin cancer risk observed in red-haired individuals.

Reactive oxygen species are normally neutralized by antioxidants such as glutathione. Polymorphisms in genes encoding **glutathione S-transferase** are associated with a twofold increased risk of actinic keratosis (AK), squamous cell carcinoma (SCC), and basal cell carcinoma (BCC). This risk escalates to sixfold in individuals with high sun exposure and reaches 12-fold in sun-exposed transplant recipients.

DNA Repair and Apoptosis

Xeroderma pigmentosum (XP) (see Chap. 109) exemplifies the link between DNA repair capacity and skin cancer risk. Mutations in any of the XP genes impair nucleotide excision repair (NER), and the clinical severity—measured by tumor burden and life expectancy—correlates directly with residual DNA repair function.

This relationship extends to the general population. Patients with actinic keratosis exhibit 30% to 50% reduced excision repair capacity in normal fibroblasts or lymphocytes. Similarly, cells from individuals with basal cell carcinoma (BCC) show diminished excision repair compared to those from healthy controls. These findings indicate that even subtle variations in DNA repair efficiency can significantly influence an individual's susceptibility to skin cancer.